
BY 518: Disease Biology

Course Code	BY 518
Course Name	Disease Biology
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	B. Tech. 3rd and 4th year, MS/M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Infectious diseases**

Bacterial: Tuberculosis, Urinary tract infection (E. coli), Pneumonia (S. aureus), Typhoid, Gut infection (H. Pylori); Viral diseases: Cough and cold (influenza), Hepatitis, Measles, Mumps, Japanese encephalitis, Polio, HIV; Parasitic diseases: E. histolytica, Leishmaniasis, Malaria, Ascaris, Giardiasis, Filariasis, Cysticercosis [10 Lectures]

- **Autoimmune diseases**

Arthritis, Lupus, Atherosclerosis, Alzheimer [8 Lectures]

- **Metabolic diseases**

Diabetes, Obesity, Cancer [12 Lectures]

- **Genetic diseases**

Haemophilia, Down syndrome, Angelman syndrome, Colour blindness [12 Lectures]

Text and Reference Books

1. Nicholas Sperelaki, **Cell Physiology Source Book: A Molecular Approach**, 4th Edition.
2. Jonathan Phillips, Paul G. Murray, Paul Kirk (Editors), **The Biology of Disease**, 3rd edition. Wiley Publications.
3. Strachan, Tom, and Andrew P. Read, **Human Molecular Genetics**, 2nd Edition, John-Wiley & Sons.
4. Dennis J. Selkoe, Eckhard Mandelkow, David M. Holtzman, **he Biology of Alzheimer Disease**, ISBN 978-1-936113-44-6.
5. Additional handouts and references from peer reviewed publications will be provided.